Oussama Darouez

Embedded Systems Engineer

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Summary

Embedded Systems Engineer with hands-on experience designing firmware, real-time systems, and embedded networking protocols industrial and IoT projects. Skilled in bare-metal and RTOS-based development on various microcontrollers, driver design and CI/CD automation for production-grade firmware.

Education

National Institute of Applied Science and Technology (INSAT) *Sept 2021 – Sept 2026*
Engineering Degree, Industrial Computing and Automation

Experience

Embedded Systems Engineering Intern *Meroux, France*
Nicolaudie Group – LightingSoft AG *March 2026 – Present*

- Developed a scalable **Modbus TCP/RS485 server** for STM32 and NXP LPC, supporting multiple concurrent clients.
- Integrated **Modbus triggers** into the show engineand the **C++ Backend** for real-time, dynamic scene parameter control.
- Built a robust **C++ Modbus client** with automated stress tests validating protocol reliability.
- Enhanced **Qt application** with Modbus configuration and import/export capabilities.
- Authored comprehensive **QTest** suites, strengthening regression coverage and system reliability.
- Migrated **Jenkins CI/CD** pipelines from Qt5 to Qt6, resolving automated test-build issues.

Technologies: C · C++ · Qt · Qtest · STM32 · NXP LPC · Wireshark · Jenkins · SEGGER Ozone · FreeROS · Lizard Analysis

Embedded Systems Engineer *Tunis, Tunisia*
MareCustos *June 2024 – Sept 2025*

- Built a **dual-slot OTA** firmware update system over Ethernet/USB with automatic fallback recovery.
- Prototyped reusable peripheral drivers for **Timers, Ethernet, Flash, ADC, and PCNT** modules.
- Designed custom **micro-ROS transport layers** for STM32, enabling reliable inter-device communication.
- Automated firmware build and testing workflows using **CI/CD** pipelines with GitHub Actions.
- Designed and debugged a custom **bootloader** with reset-cause handling for production robustness.

Technologies: C · C++ · STM32 · ESP32 · · ROS2 · micro-ROS · Unity C · GitHub Actions · Jira

Mechatronics Engineer *Tunis, Tunisia*
Orange Tunisie *May 2024 – July 2024*

- Optimized **FreeRTOS** task scheduling on ESP32 for reliable multitasking and efficient resource use.
- Restructured the **UART/Ymodem** interface for reliable command reception and task triggering.
- Established a robust **test bench** to streamline robot functionality validation.

Technologies: ESP32 · FreeRTOS · UART · Ymodem · Bash

Robotics Instructor *Manouba, Tunisia*
Discovery Club Junior *May 2023 – May 2025*

- Delivered hands-on **PCB design** training with KiCad, from schematic to prototype.
- Guided students through **digital logic design** and ATmega328P register-level programming.
- Assisted students with mechanical design and robot mechanics brainstorming using SolidWorks.

Technologies: KiCad · SolidWorks · ATmega328P · Robotics

Projects

OTA-Enabled Smart Home Devices *08/2025 – 09/2025*

- Prototyped IoT devices with **HTTPS OTA** updates and SmartConfig Wi-Fi provisioning.
- Implemented **non-volatile storage** management for reliable device parameter retrieval.
- Automated firmware build, flash, and deployment using **Bash** and **GitHub Actions** for ESP-IDF.

Technologies: ESP-IDF · Wi-Fi · GitHub Actions · Bash

Bare-Metal Drivers *01/2024 – 05/2024*

- Developed low-level **bare-metal drivers** for STM32, PIC and Atmega: GPIO, ADC, Timer, and EXTI peripherals.
- Implemented LittleFS and FlashDB to gain hands-on experience with embedded file systems and persistent storage.
- Developed **SPI/I2C** accelerometer drivers and a **PWM/ADC** brushless motor controller.

Technologies: STM32F4 · PIC16F · Atmega328P· SPI · I2C · LittleFS · FlashDB

Autonomous Robots, Eurobot *09/2022 – 05/2023*

- Engineered servo drivers and **GPIO HATs** with KiCad, validated through real-world testing.
- Implemented **PID motor control** and odometry on STM32 for precise robot movement.
- Set up **ROS1** communication between Raspberry Pi and STM32 across multi-board systems.

Technologies: STM32 · ROS1 · KiCad · 3D Printing

Skills

Microcontrollers: STM32, ESP32, PIC, ATmega328P, Raspberry Pi, NXP LPC.
Protocols: Ethernet, TCP/IP, UART, SPI, I2C, USB-CDC, CAN, Modbus, RS485, DMX, Wi-Fi.
RTOS & Tools: FreeRTOS, Qt, SEGGER Ozone, Wireshark, KiCad, SolidWorks.
Languages & CI/CD: C, C++, Python, Java, Bash, Git, GitHub Actions, Jenkins, Docker.

Languages

English: Proficient, **French**: Proficient, **Arabic**: Native.